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Reg. No: | | | | | | | | | | | |

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccreditedby NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE / M.TECH - CSE

20CS215 – CLOUD COMPUTING

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO1 Choose the false statement about cloud computing.**
   a) Private cloud doesn't employ the same level of virtualization
   b) Data center operates under average loads
   c) Private cloud doesn't pooling of resources that a cloud computing provider can achieve
   d) Abstraction enables the key benefit of cloud computing: shared, ubiquitous access

2. **CO3 A service level agreement (SLA) between two units within an organization does not ________.**
   a) Outline the strategic goals of the organization
   b) Outline the responsibilities of each party
   c) Establish the nature of the service provided
   d) Determine the transfer price of the service

3. **CO2 The technology used to distribute service to resources is referred to as ______.**
   a) load performing b) load scheduling c) load balancing d) load shifting

4. **CO2 Which AWS service is used to launch and manage virtual servers in the cloud?**
   a) Amazon S3 b) AWS Lambda c) Amazon EC2 d) Amazon RDS

5. **CO2** **Which of the following mechanisms are contained by Cloud API for accessing cloud services?**

a) Abstraction b) Replication c) Authentication d) Root Permission

6. **CO2** **Which of the following is the most important area of concern in cloud computing?**

a) Scalability b) Storage c) Security d) Elasticity

7. **CO4** **Identify the core component of Docker.**

a) Docker CLI b) Docker Engine c) Docker Server d) Docker Hypervisor

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO1** Applications and services that run on a distributed network using virtualized resources is known as ______.
9. **CO1** Elastic Compute Cloud is an example of _______ service model.
10. **CO4** _____ defines the instructions to build a Docker image.

PART - B **(5x 3 = 15 Marks)**

11. **CO1** Summarize the characteristics of Cloud Computing paradigm.
12. **CO1** Distinguish public and private clouds.
13. **CO2** How OS virtualization implemented.
14. **CO2** List out the challenges in Identity and Access Management.
15. **CO4** Differentiate containers and virtual machines.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

16. **CO2** i) Discuss about the levels of virtualization and its implementation. (8)

CO2 ii) "Virtualization is critical for cloud computing". Justify. (7)

17. **CO1** Discuss the various cloud computing services with neat sketch. (15)

18. **CO1** i) Illustrate the NIST cloud computing reference architecture with its components. (8)

CO3 ii) Develop a Service Level Agreement for banking system services with suitable performance metrics. (7)

19. **CO2** i) Describe the Xen Architecture. (8)

CO2 ii) Explain about I/O devices virtualization. (7)

20. **CO2** i) Explain the cloud services provided by Windows Azure. (8)

CO2 ii) Explain about the authentication, access management and control in cloud computing. (7)

21. **CO2** Explain the fundamental functions which are required for secure cloud computing. (15)

22. **CO4** i) Explain the components of Docker. (8)

CO4 ii) Demonstrate the Docker networking commands. (7)

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